

End-to-End Process · Idea-to-Market (I2M)

PCF 2.1 Executive Steering Committee · PCF 2.2 R&D / Innovation · PCF 2.3.1–2.3.3 Engineering, Marketing & Operations

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View 2 — End-to-End Process Specification

I2M End-to-End Process Specification · People · Process · Data · Technology

Step · PCF	Pattern · M3%	People	Process	Data · CRUD	Technology
PCF 2.1 · Executive Steering Committee					
Manage Product & Service Portfolio 2.1.1	Decision 80%	Executive Steering Committee <i>Parts/accessories variant routes to Product Management — see note</i>	Revisit portfolio against market opportunities; prioritize, retire, and align new/existing offerings to business strategy	C Product/Service Portfolio Record <i>PLM / Portfolio Mgmt Tool</i> R Market & Performance Data <i>BI / Market Intelligence Platform</i>	PLM Platform — portfolio governance AI Scenario Analysis — market-gap modeling
Manage Product & Service Life Cycle 2.1.2	Decision 80%	Executive Steering Committee	Govern introduction, launch, and retirement of products/services; commit post-launch review findings	C Lifecycle Decision Record <i>PLM / Portfolio Mgmt Tool</i> R Product/Service Portfolio Record <i>PLM / Portfolio Mgmt Tool</i>	PLM Platform — lifecycle tracking AI Performance Modeling — retirement/launch timing
PCF 2.2 · R&D / Innovation					
Perform Discovery Research 2.2.1	Knowledge 65%	R&D / Innovation Team	Coordinate early-stage R&D to identify new technologies (incl. AI) closing gaps between current offerings and market expectations	C Technology Discovery Report <i>R&D Knowledge Platform</i> R Market & Performance Data <i>BI / Market Intelligence Platform</i>	R&D Knowledge Platform — technology scouting AI Research Synthesis — trend & patent scanning
Generate New Product/Service Concepts 2.2.2	Knowledge 65%	R&D / Innovation Team	Gather, analyze, and evaluate new product/service ideas; formulate concepts	C Concept Record <i>Idea Management Platform</i> R Technology Discovery Report <i>R&D Knowledge Platform</i>	Idea Management Platform — concept pipeline AI Ideation Support — concept generation & clustering
Define Product/Service Development Requirements 2.2.3	Knowledge 65%	R&D / Innovation, Supply Chain Liaison	Identify and capture functional, interoperability, safety, and security requirements; confirm supply chain feasibility	C Development Requirements Record <i>Idea Management Platform</i> R Concept Record <i>Idea Management Platform</i>	Requirements Management Tool — requirements traceability AI Feasibility Synthesis — supply-chain constraint flagging
PCF 2.3.1 · Engineering / Design					
Design & Prototype Products/Services Gate 2.3.1	Knowledge 65%	Engineering / Design Team <i>Gated by Stage-Gate Committee approval (inline marker, not a lane — excluded from aggregate H% math)</i>	Develop design specs, build prototypes/proof-of-concepts, and test feasibility; route through Stage-Gate Committee sign-off before proceeding	C Design Specification Record <i>PLM / CAD Platform</i> R Development Requirements Record <i>Idea Management Platform</i>	CAD / PLM Platform — design & prototyping AI Generative Design — prototype iteration support
PCF 2.3.2 · Marketing					
Test Market for New/Revised Products & Services 2.3.2	Document 40%	Marketing Team	Prepare market study, conduct customer tests/interviews, and finalize product/service characteristics and business case	C Market Test Report <i>Market Research Platform</i> R Design Specification Record	Market Research Platform — customer testing

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				PLM / CAD Platform	AI Synthesis — market-study & business-case drafting
PCF 2.3.3 · Operations					
Prepare for Production/Service Delivery 2.3.3	Transaction 35%	Operations Team	Obtain capabilities/materials/equipment, request engineering/process changes, install and validate the production/delivery process	C Production Readiness Record PLM / MES Platform R Design Specification Record PLM / CAD Platform	MES Platform — production-readiness checklisting (mature non-AI automation baseline) AI Change Routing — engineering-change triage

CRUD Legend: C Create R Read U Update D Delete · No operation

CRUD MATRIX — EXECUTIVE STEERING COMMITTEE (PCF 2.1)

PCF 2.1 · Executive Steering Committee													
Data Entity	System of Record	2.1.1	2.1.2										
Product/Service Portfolio Record	PLM / Portfolio Mgmt Tool	C	R	-	-	-	-	-	-	-	-	-	-
Lifecycle Decision Record	PLM / Portfolio Mgmt Tool	-	C	-	-	-	-	-	-	-	-	-	-

Development Requirements Record (read here) is created in the R&D / Innovation matrix.

CRUD MATRIX — R&D / INNOVATION (PCF 2.2)

PCF 2.2 · R&D / Innovation													
Data Entity	System of Record	2.2.1	2.2.2	2.2.3									
Technology Discovery Report	R&D Knowledge Platform	C	R	-	-	-	-	-	-	-	-	-	-
Concept Record	Idea Management Platform	-	C	R	-	-	-	-	-	-	-	-	-
Development Requirements Record	Idea Management Platform	-	-	C	-	-	-	-	-	-	-	-	-

Design Specification Record (read here) is created in the Engineering / Design matrix; consumed downstream by Marketing and Operations.

CRUD MATRIX — ENGINEERING / DESIGN (PCF 2.3.1)

PCF 2.3.1 · Engineering / Design													
Data Entity	System of Record	2.3.1											
Design Specification Record	PLM / CAD Platform	C	-	-	-	-	-	-	-	-	-	-	-

CRUD MATRIX — MARKETING (PCF 2.3.2)

PCF 2.3.2 · Marketing													
Data Entity	System of Record	2.3.2											
Market Test Report	Market Research Platform	C	-	-	-	-	-	-	-	-	-	-	-

CRUD MATRIX — OPERATIONS (PCF 2.3.3)

PCF 2.3.3 · Operations													
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Data Entity	System of Record	2.3.3											
Production Readiness Record	PLM / MES Platform	C											

Methodology: H% values represent the proportion of human involvement at each HITL maturity level. M3 = HITL Design Intent — the designed steady-state human/AI balance for a mature implementation. M1 = Undesigned baseline. Pattern types (Decision, Knowledge, Document, Transaction, Exception) are derived from APQC PCF v8.0 SIPOC reference architecture. The inline Stage-Gate Committee approval on 2.3.1 is a decision checkpoint with no ongoing work of its own — it is marked on the step rather than given its own lane and is excluded from the aggregate H% calculation. CRUD operations reflect net-new data interactions.